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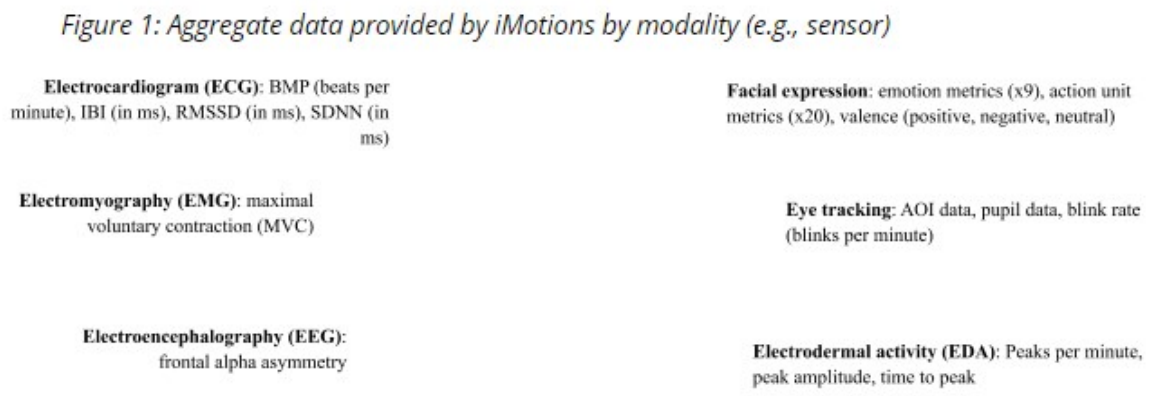
Introduction to Databasing

Introduction to Databasing

Organizing your sensor data within a database

A database is a collection of data that will allow you to have a methodical way to organize and access a large amount of information across multiple data sources. With an established database to store all your iMotions aggregate data (Figure 1) cross-linked with relevant outcome measures, such as key performance indices, you will have an efficient way to access this information and generate knowledge and insights for your group. Databasing can be categorized into “flat” and “relational” databases where flat databases are simpler versions that can be created and accessed using software like Microsoft Excel and statistical analysis programs like SPSS, SAS, and JMP. Relational databases store data in terms of complex relationship networks that require some programming knowledge to query and extract information effectively. In this knowledge document, we will focus on simple flat databases organized and stored using Microsoft Excel. If you believe relational databases are better fitted for your needs, please reach out to us to discuss potential solutions.

Figure 1: Aggregate data provided by iMotions by modality (e.g., sensor)



Considering the appropriate level of aggregation

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In this document, we provide some examples of different databases stored in Microsoft Excel for typical levels of aggregation, such as aggregation by participant or by stimulus. In these databases, each *row* will hold a set of data entries for each participant or stimulus, while the columns represent these data entries where each column holds entries from a unique data source, such as aggregated electrodermal activity peaks-per-minute data exported from iMotions or a performance outcome measure. Depending on your research question, you would want to aggregate your data accordingly either by participant or by stimulus. For the highest data fidelity, you could also have a database containing data organized in a stimulus by participant manner (Table 1) and calculate the specific aggregations (i.e., by participant across stimuli; or by stimulus across participants) whenever you are conducting your analyses. If your research questions involve retaining the time series nature of biophysiological data, please reach out to us to discuss potential solutions for aggregating and maintaining a database of sensor data by time. In addition, we can also consult with you on potential analyses that you can conduct depending on the level of aggregation that makes sense for your research questions.

View the attachments below to see the example tables and excel.

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